SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Manipulation (Level 2)	Total Marks: 50
Question Count: 5	

Code of Conduct

I _R.AKSHAYA (192524118)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R. Akshaya

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	**20%**	**Interprets all technical requirements accurately**	**Interprets most requirements correctly**	**Partial interpretation**	**Misinterprets the problem**
Analytical Reasoning	**25%**	**Strong reasoning with clear cause-effect analysis**	**Good reasoning with minor gaps**	**Basic analysis only**	**Weak or no analysis**
Method Selection	**20%**	**Chooses highly appropriate virtualization and security concepts**	**Chooses mostly suitable concepts**	**Partially suitable concepts**	**Inappropriate approach**

Solution Quality	**20%**	**Provides feasible and technically sound solution**	**Reasonable solution with minor issues**	**Basic solution with limitations**	**Unclear or invalid solution**
Presentation of Analysis	**15%**	**Well-structured and technically precise**	**Generally clear**	**Somewhat unclear**	**Difficult to follow**

Total Marks Awarded:

Assessment Tasks

Analytical Problem Solving

1. **Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.**

Feature	Type-1 Hypervisor (Bare Metal)	Type-2 Hypervisor (Hosted)
Installation	Runs directly on hardware	Runs on top of an operating system
Performance	Very high	Lower due to OS overhead
Security	Stronger because fewer attack points	Less secure since host OS can be attacked
Resource Usage	Efficient resource utilization	Additional resources used by host OS
Reliability	High availability and stability	Depends on host operating system
Manageability	Centralized management for many VMs	Better suited for individual systems
Examples	VMware ESXi, Microsoft Hyper-V, Xen	VMware Workstation, VirtualBox

Analysis

- **Performance:** Type-1 hypervisors access hardware directly, providing faster CPU, memory, and storage performance.
- **Security:** Since no host operating system exists between hardware and hypervisor, the attack surface is smaller.
- **Manageability:** Supports centralized VM management, live migration, snapshots, clustering, and high availability.

Justification

For mission-critical cloud applications, Type-1 Hypervisor is the better choice because it offers:

- Higher performance
- Better security
- Greater reliability
- Easier large-scale management

2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

A cloud provider hosts:

- Finance applications
- Healthcare applications
- Student portals

on the same physical server.

How Virtualization Supports Isolation

1. **Separate Virtual Machines**
   - Each workload runs inside its own VM.
   - Memory, CPU, and storage are isolated.
2. **Hypervisor Isolation**
   - The hypervisor prevents one VM from accessing another VM's resources.
3. **Virtual Networking**

- Virtual LANs (VLANs) and virtual switches separate network traffic.

4. **Access Control**
   - Different security policies are applied to each VM.

Benefits

- Prevents data leakage.
- Improves security.
- Enables resource sharing without interference.

Two Major Attack Surfaces:

1. Hypervisor Attack

- Attackers exploit vulnerabilities in the hypervisor, which is the software responsible for managing all virtual machines running on the physical server.
- If the hypervisor is successfully compromised, attackers may gain unauthorized access to multiple virtual machines and the sensitive data they contain.
- Regular security updates, vulnerability patching, and continuous monitoring are essential to protect the hypervisor from such attacks.

2. VM Escape

- VM escape occurs when malicious software breaks out of a virtual machine and reaches the hypervisor or another virtual machine.
- This allows attackers to bypass the isolation provided by virtualization and access confidential data or interfere with other workloads.
- Using secure hypervisors, applying security patches, and restricting unnecessary VM privileges help reduce the risk of VM escape attacks.

Conclusion

Virtualization provides strong isolation, but securing the hypervisor and preventing VM escape are critical.

3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%,storage I/O latency 35ms, average VM boot time 170 s. Explain the likely bottlenecks.

Given Snapshot

- CPU Utilization = 92%
- Memory Utilization = 88%
- Storage I/O Latency = 35 ms
- Average VM Boot Time = 170 seconds

Analysis

CPU (92%)

- Nearly fully utilized.
- Indicates CPU bottleneck.
- VMs compete for processor time.

Memory (88%)

- Very high memory usage.
- Possible memory contention or swapping.

Storage Latency (35 ms)

- Storage response is slow.
- Normal latency is usually below 10 ms.
- Indicates disk bottleneck.

VM Boot Time (170 s)

- Much higher than expected.
- Caused by CPU and storage delays.

Likely Bottlenecks

- CPU saturation
- Memory pressure
- Slow storage subsystem

Corrective Actions

1. Add more CPU resources.
2. Increase physical RAM.
3. Upgrade HDD to SSD/NVMe storage.
4. Use storage caching.
5. Balance workloads across additional hosts.
6. Remove idle or unused VMs.
7. Enable resource monitoring and autoscaling.

Conclusion

The host is overloaded in CPU, memory, and storage, leading to poor VM performance and long boot times.

4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

Traditional Datacenter

- Hardware configured manually.

- Dedicated servers.
- Fixed storage.
- Physical network devices.
- Slow deployment.

Software Defined Datacenter
All infrastructure is controlled through software.

Compute Virtualization
- Physical servers are divided into multiple virtual machines.
- Rapid provisioning and migration.
- Better resource utilization.

Storage Virtualization
- Combines multiple storage devices into a single storage pool.
- Dynamic allocation.
- Easier backup and disaster recovery.

Network Virtualization
- Uses virtual switches and software-defined networking (SDN).
- Quick network configuration.
- Better traffic management.

Agility Improvements

Traditional Datacenter	Software Defined Datacenter
Manual deployment	Automated deployment
Hardware dependent	Software controlled
Slow scaling	Rapid scaling
Fixed resources	Dynamic resource allocation
High operational effort	Automated management

Conclusion

SDDC improves agility by:

- Automating infrastructure
- Enabling rapid provisioning
- Simplifying management
- Improving scalability and resource utilization

5. **A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.**

A multi-cloud environment connects multiple cloud providers.
Failure occurs because:

- Different identity systems
- Different authentication methods
- Different user roles
- No common trust relationship

As a result:

- Users cannot access services across clouds.
- Authentication failures occur.

Secure Identity and Privacy Design

1. Federated Identity Management

- Users authenticate once.
- Identity is trusted across cloud providers.

2. Single Sign-On (SSO)

- Users log in once and access multiple cloud services.

3. Identity Provider (IdP)

- Central authentication server verifies users.
- Cloud providers trust the IdP.

4. Standard Authentication Protocols

Use standards such as:

- SAML
- OAuth 2.0
- OpenID Connect

5. Role-Based Access Control (RBAC)

- Users receive permissions based on their roles.
- Prevents unauthorized access.

6. Privacy Protection

- Encrypt identity information.
- Share only necessary user attributes.
- Use Multi-Factor Authentication (MFA).
- Maintain audit logs for monitoring.

Benefits

- Secure authentication
- Improved privacy

- Cross-cloud interoperability
- Reduced password management
- Easier user access

Conclusion

Using federated identity, SSO, standard authentication protocols, RBAC, MFA, and encrypted identity data enables secure access across multiple cloud providers while protecting user privacy.